

```

#include <LiquidCrystal.h>

LiquidCrystal lcd(13,12,11,10,9,8);

#define R_PHASE A0
#define Y_PHASE A1
#define B_PHASE A2

#define RED_LED 2
#define YELLOW_LED 3
#define BLUE_LED 4

#define BUZZER 5

float maxDistance = 4.0;

void setup()
{
    lcd.begin(16,2);

    pinMode(RED_LED,OUTPUT);
    pinMode(YELLOW_LED,OUTPUT);
    pinMode(BLUE_LED,OUTPUT);

    pinMode(BUZZER,OUTPUT);

    lcd.clear();
    lcd.print("UNDERGROUND");
    lcd.setCursor(0,1);
    lcd.print("CABLE FAULT");
    delay(2000);
}

void loop()
{
    checkPhase(R_PHASE, "R+", RED_LED);

    delay(1500);

    checkPhase(Y_PHASE, "Y+", YELLOW_LED);

    delay(1500);

    checkPhase(B_PHASE, "B+", BLUE_LED);

    delay(1500);
}

void checkPhase(int analogPin,String phase,int ledPin)
{

```

```

digitalWrite(REDA_LED,LOW);
digitalWrite(YELLOW_LED,LOW);
digitalWrite(BLUE_LED,LOW);

digitalWrite(ledPin,HIGH);

int adc = analogRead(analogPin);

float voltage = adc * 5.0 / 1023.0;

lcd.clear();

lcd.setCursor(0,0);
lcd.print(phase);

if(voltage >= 4.8)
{
    lcd.print(" NF");

    digitalWrite(BUZZER,LOW);

    lcd.setCursor(0,1);
    lcd.print("No Fault");
}
else
{
    float distance;

    distance = ((5.0 - voltage)/5.0)*maxDistance;

    digitalWrite(BUZZER,HIGH);

    lcd.setCursor(0,1);
    lcd.print(distance,1);
    lcd.print(" KM Fault");
}

delay(1000);

digitalWrite(BUZZER,LOW);
}

```